

When to Rerank: Calibrated Risk- and Cost-Aware Routing for Dense–Hybrid Retrieval Cascades

Wasim Iqbal

Abstract—Static retrieval cascades often invoke Cross-Encoder reranking for every query, although reranking may be unnecessary or reduce nDCG. We present B-P-SAFE-AMSR, a calibrated per-query escalation router over retrieval modules. For each query, it estimates gain probability, retrieval-quality regression probability, expected nDCG@10 delta, and latency cost, then chooses Dense retrieval or a Deep Hybrid pipeline. We evaluate the binary action space on four BEIR datasets with three data-split seeds. On seed 42, high-recall routing improves over Dense on SciFact, FiQA, and NFCorpus while reducing measured end-to-end latency relative to always-on Deep Hybrid by 20.4%–31.6%; ArguAna exhibits a protection/no-benefit regime because always-on reranking is not reliably useful, while selective routing improves nDCG. Comparisons with static, heuristic, learned, random, and oracle routers expose dataset-dependent tradeoffs rather than universal dominance. Paired tests and multi-seed analyses quantify uncertainty. Limitations include binary routing, incomplete matched-activation and disagreement baselines, calibration diagnostics, and no production or content-safety evaluation.

Index Terms—adaptive retrieval, neural reranking, retrieval routing, calibrated classification, cost-aware inference, information retrieval

I. INTRODUCTION

Modern search and retrieval-augmented generation systems commonly use a fast first-stage retriever followed by a more expensive neural reranker [1], [2]. Uniformly applying the expensive stage is simple, but it ignores query-level variation. Some queries already have a strong Dense ranking; other queries benefit from lexical evidence, semantic expansion, and Cross-Encoder scoring. Moreover, the maximum-compute endpoint is not uniformly better: reranking can reorder an already useful list and reduce nDCG@10 for an individual query.

We study whether a router can decide, before executing the full cascade, when the extra computation is justified. We call the implementation *B-P-SAFE-AMSR* (Binary Probabilistic Safety-Aware Adaptive Multi-Source Retrieval), shortened to B-P-SAFE after this first use. The word “harm” means a decrease in retrieval quality relative to Dense retrieval. The paper studies retrieval-quality harm, not content safety harm.

This work is not a new retriever or reranker. It is a calibrated routing layer over existing retrieval modules. Its action space in the reported experiments is deliberately binary: A_0^{Dense} or $A_6^{\text{DeepHybrid}}$. The latter combines Dense and BM25 retrieval, cached semantic-neighbour expansion, reciprocal-rank fusion, and Cross-Encoder reranking. The router estimates whether this

escalation will help or hurt, its expected nDCG@10 change, and its latency.

This paper makes four contributions:

- 1) a utility formulation of neural reranking as a calibrated per-query escalation decision balancing expected retrieval gain, regression risk, latency, candidate count, and recovery probability;
- 2) a reproducible binary Dense-versus-Deep Hybrid implementation using 25 query, score-distribution, lexical-disagreement, and graph features;
- 3) a four-dataset analysis of selective-escalation and protection/no-benefit regimes using paired tests, query-level confidence intervals, and latency/quality diagnostics; and
- 4) comparisons with static, random, heuristic, learned, and oracle routers, plus sensitivity analysis over three train/validation/test split seeds.

The results should not be read as universal dominance. On seed 42, high-recall B-P-SAFE improves over Dense on SciFact, FiQA, and NFCorpus while saving 20.4%–31.6% of measured end-to-end latency relative to always-on Deep Hybrid. On ArguAna, where the mean Deep Hybrid score is below Dense, routing improves quality by escalating only selected queries. Strong simple routers remain competitive in some settings, and the multi-seed analysis exposes split sensitivity.

II. RELATED WORK

A. Neural Reranking and Retrieval Cascades

Dense retrieval provides an efficient semantic first stage [1]; Cross-Encoders can improve ranking by jointly scoring a query–document pair, but at substantially higher inference cost [2]. Multi-stage ranking has therefore long been framed as an efficiency–effectiveness problem. Operational cascade ranking [3], joint cascade optimization [4], and learned early exits for additive rankers [5] reduce work within or across ranking stages. B-P-SAFE instead routes the whole query between two already-defined retrieval endpoints using predicted query-level outcomes.

Sparse, dense, and multi-vector signals can also be unified within retrievers such as BGE-M3 [6]. Our experiment uses BGE-M3 dense embeddings, BM25 lexical evidence, semantic-neighbour expansion, and a BGE reranker; it does not propose a new representation model. Reciprocal-rank fusion supplies a score-scale-robust combination of retrieval lists [7].

B. Selective Prediction, Calibration, and Cost-Aware Dispatch

Selective prediction learns when to accept or reject a model output under a coverage–risk trade-off [8]. Probability

calibration asks whether predictive confidence is numerically meaningful [9]. These ideas motivate separate estimates of reranking gain and retrieval regression. Our implementation uses sigmoid calibration around logistic models, but the reported experiments do not establish a formal calibration or conformal guarantee.

Cost-aware dispatch appears in model cascades beyond retrieval. FrugalGPT learns how to combine language models under performance and cost objectives [10]; Adaptive-RAG selects among no-retrieval, single-step, and iterative strategies using question complexity [11]. These systems route generation or retrieval-augmented answering strategies, whereas B-P-SAFE routes retrieval endpoints and explicitly models negative nDCG@10 deltas.

Most directly, Adaptive Re-Ranking [12] studies utility-based routing among BM25, dense reranking, and heavier neural reranking. That concurrent work emphasizes the potential and difficulty of learning per-query routing. Our study differs in its binary Dense–multi-source-hybrid endpoint, explicit gain and regression probabilities, mode-dependent admissibility rules, and paired query-level evidence. The comparison is conceptual; the two systems were not run in a shared experimental harness.

III. METHOD

A. Endpoints and Problem Definition

For a query representation x , Dense retrieval is the zero-incremental-cost reference action A_0 . The escalation action A_6 retrieves 100 Dense and 200 BM25 candidates, expands the top 20 Dense seeds by two hops over cached five-neighbour lists, fuses a pool of up to 400 candidates, and reranks the top 100 with a Cross-Encoder. The output metric is nDCG@10.

Let $\hat{\delta}(a, x)$ be the predicted nDCG@10 change relative to Dense, $\Delta t(a, x)$ the predicted action latency in milliseconds, and $n_{\text{cand}}(a)$ the action candidate count. Let $P_{\text{gain}}(a | x)$ denote the predicted probability that the improvement exceeds 0.05, and $P_{\text{harm}}(a | x)$ the predicted probability that the change is below -0.01 . The full utility is

$$\begin{aligned} \mathcal{U}(a | x) = & \hat{\delta}(a, x) \\ & - \lambda_{\text{harm}} P_{\text{harm}}(a | x) - \lambda_{\text{lat}} \Delta t(a, x) \quad (1) \\ & - \lambda_{\text{cand}} n_{\text{cand}}(a) + \lambda_{\text{rec}} P_{\text{gain}}(a | x). \end{aligned}$$

Equation (1) is the general utility used to rank candidate actions. In the binary experiments in this paper, the deployed decision reduces to choosing between A_0^{Dense} and $A_6^{\text{DeepHybrid}}$. The candidate penalty is constant for the deployed A_6 configuration and is retained in Equation (1) for generality.

B. Truthful Deployment Rule

The implementation applies utility, probability gates, and documented mode-specific soft overrides. Let $\mathcal{G}_m(x)$ be true when both $P_{\text{gain}}(A_6 | x) \geq \tau_{\text{gain}}$ and $P_{\text{harm}}(A_6 | x) \leq \tau_{\text{harm}}$, or when the override for mode m applies. The deployed rule is

$$a^*(x) = \begin{cases} A_6^{\text{DeepHybrid}}, & \mathcal{U}(A_6 | x) > 0 \wedge \mathcal{G}_m(x), \\ A_0^{\text{Dense}}, & \text{otherwise.} \end{cases} \quad (2)$$

Lite has no override. Balanced admits a rejected escalation if $\hat{\delta} > 0.02$ and $P_{\text{harm}} < \tau_{\text{harm}} + 0.10$; high recall uses $\hat{\delta} > 0.005$ and $P_{\text{harm}} < \tau_{\text{harm}} + 0.20$. A candidate still needs positive utility after an override. These clauses are included because they are present in the evaluated router; omitting them would describe a different decision rule.

The source exposes an “LCB safety” flag for Lite, but its current value is the point utility itself. It is therefore only an additional non-positive-utility check, not a statistical lower confidence bound. Bootstrap or conformal utility bounds remain future work.

C. Prediction Models and Calibration Objective

All continuous features are standardized on the training split. Ridge regressions estimate $\hat{\delta}$ and latency. Logistic regressions estimate gain and harm labels with balanced class weights. When both classes have at least two observations, `CalibratedClassifierCV` uses sigmoid (Platt-style) calibration with the feasible cross-validation count; degenerate or failed fits use an empirical-prior model. Thus the method has an explicit probabilistic calibration objective, not a guarantee. The artifacts do not report expected calibration error, Brier score, reliability curves, or temporal drift, which limits how literally the probabilities should be interpreted.

Thresholds are selected on validation queries from a fixed grid to maximize realized validation nDCG@10 delta. Mode coefficients are configured before test evaluation; saved per-dataset settings are reported in Appendix C. The model never uses a test relevance label for training or tuning.

D. Routing Features

The router uses 25 features: query length, numeric and uppercase markers, average and maximum token IDF, lexical specificity; Dense maximum, mean, standard deviation, normalized entropy, and top-score gaps; analogous normalized BM25 statistics; Dense–BM25 Jaccard overlap at 10 and 50, rank correlation, candidate novelty, and source disagreement; and maximum, mean, and zero fraction of graph degrees. BM25 features are computed for routing even if the final decision remains Dense, so the reported Dense endpoint and router latency are not interchangeable cost definitions.

E. Semantic-Neighbour Expansion

Document embeddings are normalized for inner-product retrieval. We use FAISS [13] `IndexFlatIP` to compute exact top-six neighbours for every document, remove the self-match, and cache five-neighbour adjacency lists. Query-time graph expansion traverses these lists from Dense seeds. No HNSW index is constructed, and the results should not be described as HNSW or approximate-nearest-neighbour graph search. The operational method is cached top- k neighbour expansion over dense embedding space.

IV. EXPERIMENTAL DESIGN

A. Datasets, Splits, and Models

We evaluate SciFact, FiQA-2018, NFCorpus, and ArguAna from BEIR [14]. Stratified query splits allocate 40% to router

B-P-SAFE-AMSR Binary Retrieval Routing

A calibrated routing layer over existing Dense and Deep Hybrid endpoints

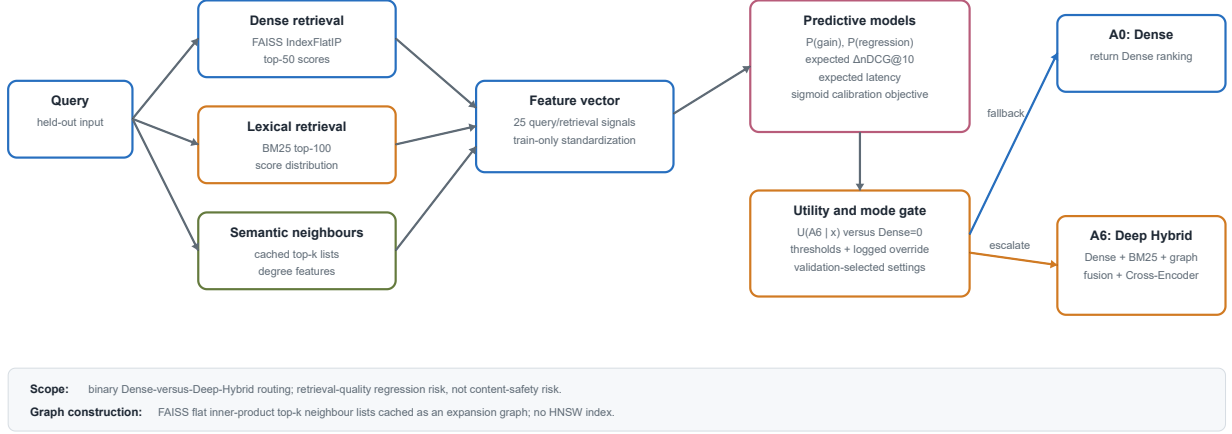


Fig. 1. Evaluated binary architecture. The router predicts gain, retrieval regression, nDCG@10 delta, and latency, then selects Dense or the predefined Deep Hybrid endpoint. Neighbour lists are computed with flat inner-product FAISS; no HNSW index is used.

training, 10% to validation, and 50% to testing. Seed 42 is the primary split; seeds 123 and 2026 repeat the split procedure. Their test sets are therefore different. Multi-seed results measure split sensitivity rather than repeated training on one fixed test set.

The seed-42 test sizes are 152, 325, 162, and 706 queries, respectively. Dense retrieval uses BAAI/bge-m3; reranking uses BAAI/bge-reranker-v2-m3. BM25 uses whitespace-lowercase tokenization. Dense and BM25 retrieve 100 and 200 documents within A_6 , the semantic graph uses five neighbours, fusion retains at most 400 candidates, and the Cross-Encoder reranks 100. Both endpoints return ten documents for evaluation.

Experiments used Python 3.11.9 with CUDA on an NVIDIA GeForce RTX 5070 Ti. Embeddings and semantic-neighbour lists were cached. Latency is measured in milliseconds on this environment and should be compared within a run, not treated as hardware-independent throughput.

B. Router Modes

Lite is deliberately conservative and selects Dense for all seed-42 test queries. Balanced and high-recall modes trade latency and recovery differently. Because validation sets can be small (e.g., 29 queries for SciFact seed 42), these operating points may vary across splits. We therefore foreground the per-seed results rather than interpreting a single threshold as a stable universal policy.

C. Baselines and Fairness

We compare the following routers on the same held-out queries:

- **Dense-only** and **Always-Hybrid**: static endpoints.

- **Random**: samples A_6 with the validation fraction for which Hybrid beat Dense by more than 0.001. It is seeded but is *not* matched to each method’s test activation.
- **Dense-margin**: escalates below a validation-selected Dense score-gap threshold.
- **Dense-entropy**: escalates above a validation-selected Dense score-entropy threshold.
- **Regression-only**: a training-split Ridge delta predictor with a validation-selected threshold.
- **Classification-only**: a training-split logistic gain classifier with a validation-selected probability threshold.
- **Oracle**: uses test per-query endpoint scores to select the better action. It is an unattainable diagnostic upper bound, not a deployable baseline.

Heuristic and learned thresholds maximize validation nDCG@10 over their defined grids; test scores are computed only after tuning. No BM25-disagreement or cost-only router has a validated result artifact, so neither is reported. Baseline-router latency combines endpoint latency selected per query but excludes uninstrumented router decision overhead; B-P-SAFE latency is measured end-to-end. This asymmetry prevents fine-grained latency superiority claims against learned/heuristic routers. Comparisons with always-on Deep Hybrid use consistently measured endpoint/action latency and are the main latency conclusion.

D. Metrics and Statistical Protocol

For system mean $\bar{q}_s$, define deltas against Dense and Deep Hybrid as

$$\begin{aligned}\Delta_D &= \bar{q}_{\text{B-P-SAFE}} - \bar{q}_{\text{Dense}}, \\ \Delta_H &= \bar{q}_{\text{B-P-SAFE}} - \bar{q}_{\text{DeepHybrid}}.\end{aligned}\tag{3}$$

Latency saving (LS), using measured mean action latency $\bar{t}$, is

$$LS = 1 - \frac{\bar{t}_{\text{B-P-SAFE}}}{\bar{t}_{\text{DeepHybrid}}}. \quad (4)$$

Hybrid gain retention (HGR) is

$$HGR = \frac{\bar{q}_{\text{B-P-SAFE}} - \bar{q}_{\text{Dense}}}{\bar{q}_{\text{DeepHybrid}} - \bar{q}_{\text{Dense}}}. \quad (5)$$

HGR is not applicable when Deep Hybrid does not improve on Dense; tables show 0 by convention and mark the regime as protection/no-benefit. Recovery capture (RC) compares the positive Hybrid gain retained by B-P-SAFE on queries labeled hard by Dense:

$$RC = \frac{\max(q_{\text{B-P-SAFE}} - q_{\text{Dense}}, 0)_{\text{hard}}}{\max(q_{\text{DeepHybrid}} - q_{\text{Dense}}, 0)_{\text{hard}}}. \quad (6)$$

Oracle gap closed (OGC) is

$$OGC = \frac{\bar{q}_{\text{B-P-SAFE}} - \bar{q}_{\text{Dense}}}{\bar{q}_{\text{oracle}} - \bar{q}_{\text{Dense}}}. \quad (7)$$

Harm avoidance (HA) is the reduction, relative to always-on Deep Hybrid, in mean negative delta on Dense-easy queries. Hybrid activation rate (HAR) is the fraction assigned to A_6 .

All statistical comparisons are paired by query ID. We report two-sided paired t tests, Wilcoxon signed-rank tests, 5,000-draw paired bootstrap intervals for the mean delta, 2,000-draw paired sign-permutation tests, and paired Cohen’s d_z where discussed. Stored test statistics were independently recomputed from per-query CSV files. Because multiple datasets and modes are tested, p -values are interpreted as evidence for paired differences within each predefined setting rather than as a single family-wise confirmatory test. We do not infer guarantees from significance tests.

V. RESULTS

A. Endpoint Trade-offs

Table I gives the seed-42 endpoints and routed operating points; Fig. 2 complements it with the high-recall mean and sample standard deviation across all three split seeds. No single operating point dominates across datasets. On SciFact seed 42, high recall achieves 0.7023 versus 0.6466 for Dense and 0.7330 for Deep Hybrid, with 31.6% latency saving and 66.4% activation. On FiQA, the same mode reaches 0.4285 versus 0.4085 and 0.4327, saving 28.0%. NFCorpus reaches 0.3626 versus 0.3339 and 0.3632, nearly retaining the endpoint mean at 20.4% saving.

ArguAna reverses the endpoint ordering: Dense is 0.3946 and Deep Hybrid is 0.3915. Balanced routing reaches 0.4069 while activating Hybrid for 15.0% of queries and saving 66.1% latency. This is a protection/no-benefit regime, not evidence that Deep Hybrid is an upper-quality reference.

B. Paired Statistical Validation

Table II verifies the Dense comparisons. SciFact high recall has $\Delta_D = +0.0557$, paired $t = 2.796$, $p = 0.00584$, and Cohen’s $d_z = 0.227$; its 95% paired-bootstrap interval is [0.0191, 0.0955]. Balanced SciFact is also positive

($p = 0.0114$). FiQA balanced is directionally positive but not significant by the paired t test ($p = 0.146$); high recall has $p = 0.0448$, while its Wilcoxon result is not significant at 0.05. NFCorpus balanced has $p_t = 0.0273$ and Wilcoxon $p = 0.0255$; high recall has $p_t = 0.0126$. On ArguAna, balanced and high recall have paired- t $p = 0.00791$ and 0.0393, respectively, but the latter’s Wilcoxon $p = 0.0530$ is borderline. These mixed tests warrant setting-specific, not universal, claims.

C. Router Baselines

Figure 4 summarizes the quality–activation landscape; full tables appear in Appendix A. Dense-margin is especially strong on SciFact, and regression-only exceeds B-P-SAFE on FiQA. On NFCorpus, always-on and classification-only Hybrid remain strong, while B-P-SAFE offers intermediate activation. On ArguAna, several selective rules outperform both static endpoints. Thus the contribution is a risk- and cost-aware formulation with explicit diagnostics, not a claim to beat every router.

D. Latency Composition

Figure 5 reports SciFact high-recall latency as a transparent example. Mean B-P-SAFE latency is 486.2 ms versus 710.6 ms for always-on Deep Hybrid. With 66.4% activation, the implied conditional Hybrid call costs approximately 704 ms. Small components and decision overhead are sensitive to measurement resolution, so the paper does not claim precise component-level speedups.

E. Three-Split Sensitivity

Table III and Fig. 6 show that all balanced and high-recall runs beat Dense in mean nDCG@10, but not every seed is statistically indistinguishable from Deep Hybrid. NFCorpus high-recall HGR exceeds 1 on average because some split-specific selective combinations outperform the corresponding endpoint means. FiQA has particularly large HAR and HGR variation. Multi-seed evaluation reduces dependence on a single train/validation/test split, but three different split seeds do not quantify repeated-training variance on an identical test set.

VI. DISCUSSION

A. What the Router Learns—and What It Does Not

The evidence supports three conclusions. First, query-level escalation can recover part of the gain of an expensive endpoint while avoiding some calls. Second, endpoint ordering is dataset-dependent, so Deep Hybrid is best treated as a maximum-compute reference and often, but not always, the higher-quality endpoint. Third, simple router baselines can be difficult to beat. Explicit harm and gain probabilities improve interpretability of the decision, but the current experiments do not isolate their causal contribution through ablation.

The ArguAna result illustrates retrieval self-protection: selective execution creates a mixture better than either mean endpoint by choosing useful Hybrid cases and retaining Dense elsewhere. It does not establish broader AI safety. Likewise, sigmoid calibration makes the model’s probabilistic objective explicit but does not prove that its probabilities remain calibrated across datasets or deployment drift.

TABLE I

SEED-42 ENDPOINT QUALITY AND MEASURED END-TO-END LATENCY TRADE-OFFS. LS IS RELATIVE TO ALWAYS-ON DEEP HYBRID; HGR IS NOT APPLICABLE WHEN HYBRID DOES NOT IMPROVE ON DENSE AND IS REPORTED AS 0.

Dataset	Mode	Dense	Hybrid	B-P-SAFE	Δ_D	Δ_H	LS	HAR	HGR
SciFact	Lite	0.6466	0.7330	0.6466	+0.0000	-0.0865	97.4%	0.0%	0.000
SciFact	Balanced	0.6466	0.7330	0.6965	+0.0499	-0.0365	34.3%	63.8%	0.577
SciFact	High recall	0.6466	0.7330	0.7023	+0.0557	-0.0308	31.6%	66.4%	0.644
FiQA	Lite	0.4085	0.4327	0.4085	+0.0000	-0.0243	79.4%	0.0%	0.000
FiQA	Balanced	0.4085	0.4327	0.4217	+0.0132	-0.0111	43.2%	45.8%	0.544
FiQA	High recall	0.4085	0.4327	0.4285	+0.0200	-0.0043	28.0%	64.0%	0.823
NFCorpus	Lite	0.3339	0.3632	0.3339	+0.0000	-0.0293	99.0%	0.0%	0.000
NFCorpus	Balanced	0.3339	0.3632	0.3579	+0.0240	-0.0053	45.6%	54.9%	0.819
NFCorpus	High recall	0.3339	0.3632	0.3626	+0.0287	-0.0006	20.4%	79.6%	0.978
ArguAna	Lite	0.3946	0.3915	0.3946	+0.0000	+0.0030	77.5%	0.0%	0.000
ArguAna	Balanced	0.3946	0.3915	0.4069	+0.0123	+0.0154	66.1%	15.0%	0.000
ArguAna	High recall	0.3946	0.3915	0.4064	+0.0118	+0.0148	60.3%	22.8%	0.000

Quality Gain and End-to-End Latency Saving

High-recall mode; mean $\pm$ sample SD across data-split seeds 42, 123, and 2026

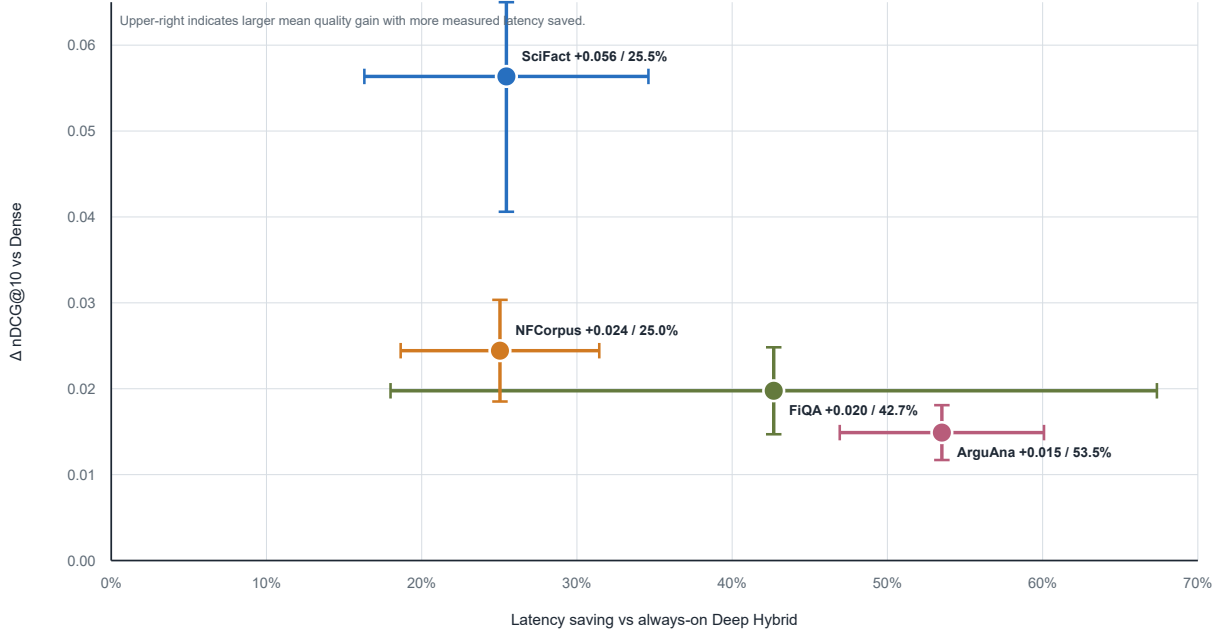


Fig. 2. High-recall mean quality gain over Dense versus mean measured end-to-end latency saving over three split seeds. Error bars are sample standard deviations. ArguAna’s positive routed gain coexists with a no-benefit always-on Hybrid endpoint.

B. Limitations

Binary action space. Only Dense and one fixed Deep Hybrid configuration are validated. Although the code enumerates additional actions, multi-action routing is future work.

Baseline gaps and latency asymmetry. Random activation is validation-derived rather than matched to each test policy. No validated BM25-disagreement, cost-only, or direct competitor reproduction is available. Baseline decision overhead is uninstrumented, whereas B-P-SAFE is measured end-to-end. Therefore only endpoint-relative latency saving is emphasized.

Calibration and uncertainty. We do not report Brier score, expected calibration error, reliability diagrams, confidence-stratified risk, or drift tests. The code’s LCB flag is not an implemented lower confidence bound. A journal study

should add out-of-fold calibration diagnostics and conformal or bootstrap decision uncertainty.

Scope and repeated evidence. Four BEIR datasets and three data-split seeds do not span all domains. Split seeds change the test queries and do not replace repeated training on a fixed test set. Significance tests are uncorrected across the reported family and are treated as descriptive setting-level evidence.

Systems scope. Latency depends on one GPU, local caches, batching, and software versions. Some subcomponent times are instrumentation estimates. There is no online production deployment, concurrency/load study, energy measurement, content-safety evaluation, or downstream generation evaluation.

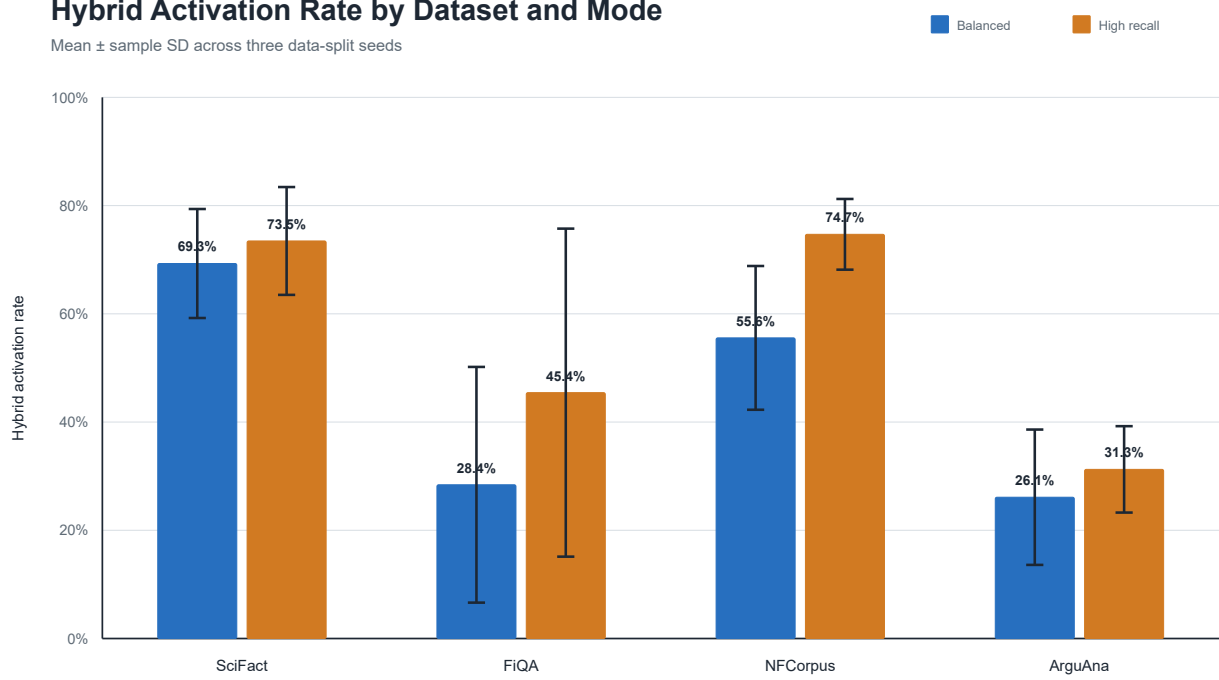


Fig. 3. Mean hybrid activation for balanced and high-recall modes across three split seeds; error bars are sample standard deviations. Activation is a policy outcome, not a fixed budget.

TABLE II

SEED-42 PAIRED VALIDATION OF B-P-SAFE AGAINST DENSE. ALL TESTS USE THE SAME HELD-OUT QUERIES; CIs AND PERMUTATIONS USE 5,000 AND 2,000 DRAWS, RESPECTIVELY.

Dataset	Mode	n	Δ_D	t	p_t	W	p_W	95% bootstrap CI	p_{perm}
SciFact	Balanced	152	+0.0499	2.563	0.011	252.0	0.021	[+0.0139, +0.0887]	0.007
SciFact	High recall	152	+0.0557	2.796	0.006	260.0	0.010	[+0.0191, +0.0955]	0.004
FiQA	Balanced	325	+0.0132	1.456	0.146	1298.0	0.227	[−0.0041, +0.0318]	0.156
FiQA	High recall	325	+0.0200	2.014	0.045	2340.5	0.088	[+0.0011, +0.0401]	0.048
NFCorpus	Balanced	162	+0.0240	2.227	0.027	706.0	0.026	[+0.0042, +0.0467]	0.029
NFCorpus	High recall	162	+0.0287	2.522	0.013	1292.0	0.008	[+0.0071, +0.0520]	0.010
ArguAna	Balanced	706	+0.0123	2.664	0.008	1182.5	0.007	[+0.0033, +0.0216]	0.005
ArguAna	High recall	706	+0.0118	2.065	0.039	3425.0	0.053	[+0.0003, +0.0230]	0.034

VII. REPRODUCIBILITY STATEMENT

The submission accompanies source, vector and raster figures, generated tables, query-level evidence, and an independent audit script. The canonical evidence directory is `results/validated`; each run stores per-query endpoint and routed scores, baseline decisions, latency, thresholds, split sizes/hash, Git commit, environment, and hardware metadata. The table auditor verifies required files, unique and matched query IDs, means, paired statistics, bootstrap intervals, and sign-permutation results across 36 dataset–seed–mode runs. Figure generation reads the same validated artifacts. All manuscript tables are generated rather than manually transcribed.

The primary environment used Python 3.11.9, BGE-M3 embeddings, BGE reranker v2-M3, FAISS flat inner-product search, CUDA, and an NVIDIA GeForce RTX 5070 Ti. Seeds are 42, 123, and 2026. Cached embeddings are used; reproducing from scratch requires downloading the public

models and BEIR datasets. Exact commands and paths are listed in the submission checklist.

VIII. CONCLUSION

B-P-SAFE casts expensive neural reranking as a per-query decision under predicted quality, retrieval-regression risk, latency, and candidate cost. Across four BEIR datasets, selective escalation improves on Dense in the reported balanced/high-recall runs and reduces measured latency relative to always-on Deep Hybrid, while ArguAna shows why maximum compute cannot be assumed to maximize mean retrieval quality. The results also show that routing remains difficult: simple thresholds can be competitive, operating points vary across splits, and current calibration diagnostics are incomplete. The appropriate conclusion is therefore a defensible quality–latency–risk framework and evidence base, not a universal superiority or safety guarantee.

Selected Router Baselines: Quality–Latency Trade-offs

Seed 42; dots show latency saving and labels show Δ nDCG@10 versus Dense; B-P-SAFE uses high-recall mode



Latency saving vs always-on Deep Hybrid

Fig. 4. Seed-42 router baseline comparison. Values show nDCG@10 and hybrid activation; the oracle is a test-label upper bound. Strongest practical methods vary by dataset.

TABLE III

THREE-SPLIT SENSITIVITY ANALYSIS OVER SEEDS 42, 123, AND 2026 (MEAN $\pm$ SAMPLE SD). COUNTS REPORT SEEDS BEATING DENSE AND SEEDS WITH NO SIGNIFICANT NEGATIVE DIFFERENCE FROM DEEP HYBRID.

Dataset	Mode	nDCG@10	LS (%)	HAR (%)	HGR	OGC	RC	HA	$> D$	$\not\prec_{\text{sig}} H$
SciFact	Balanced	0.700 ± 0.034	29.7 ± 9.3	69.3 ± 10.1	0.704 ± 0.116	0.443 ± 0.092	0.742 ± 0.105	0.004 ± 0.004	3/3	2/3
SciFact	High recall	0.705 ± 0.033	25.5 ± 9.2	73.5 ± 10.0	0.783 ± 0.121	0.490 ± 0.072	0.806 ± 0.124	0.004 ± 0.004	3/3	2/3
FiQA	Balanced	0.422 ± 0.002	56.4 ± 17.5	28.4 ± 21.8	0.478 ± 0.308	0.145 ± 0.073	0.360 ± 0.247	0.059 ± 0.011	3/3	2/3
FiQA	High recall	0.431 ± 0.003	42.7 ± 24.7	45.4 ± 30.3	0.836 ± 0.457	0.261 ± 0.093	0.591 ± 0.302	0.049 ± 0.012	3/3	2/3
NFCorpus	Balanced	0.344 ± 0.014	43.8 ± 13.3	55.6 ± 13.3	0.823 ± 0.004	0.349 ± 0.077	0.702 ± 0.147	0.023 ± 0.013	3/3	3/3
NFCorpus	High recall	0.349 ± 0.014	25.0 ± 6.4	74.7 ± 6.5	1.067 ± 0.108	0.446 ± 0.062	0.873 ± 0.077	0.016 ± 0.012	3/3	3/3
ArguAna	Balanced	0.411 ± 0.007	57.6 ± 10.0	26.1 ± 12.5	0.000 ± 0.000	0.122 ± 0.032	0.328 ± 0.135	0.163 ± 0.027	3/3	3/3
ArguAna	High recall	0.411 ± 0.007	53.5 ± 6.6	31.3 ± 8.0	0.000 ± 0.000	0.129 ± 0.033	0.383 ± 0.109	0.157 ± 0.017	3/3	3/3

Per-Query Latency Components

SciFact, seed 42, high-recall mode; Cross-Encoder mean includes zero for non-invoked queries

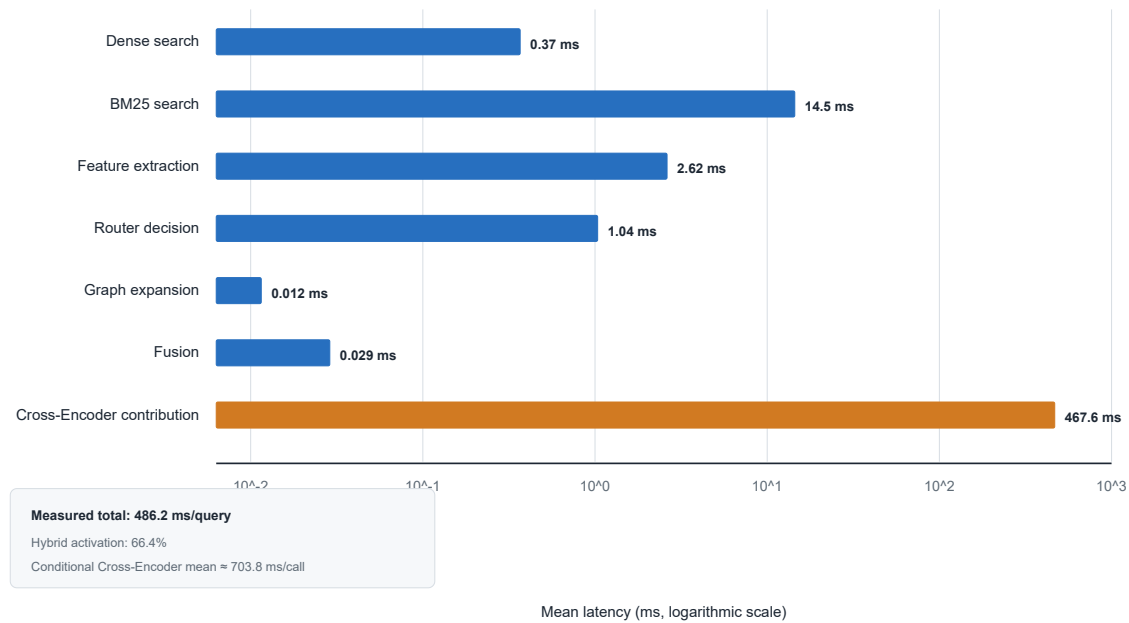


Fig. 5. Measured seed-42 SciFact high-recall latency. The dominant cost is conditional Cross-Encoder execution; component values are diagnostic measurements, not portable hardware constants.

High-Recall Sensitivity Across Data-Split Seeds

Per-seed nDCG@10 and hybrid activation; lines connect only repeated split seeds within each dataset

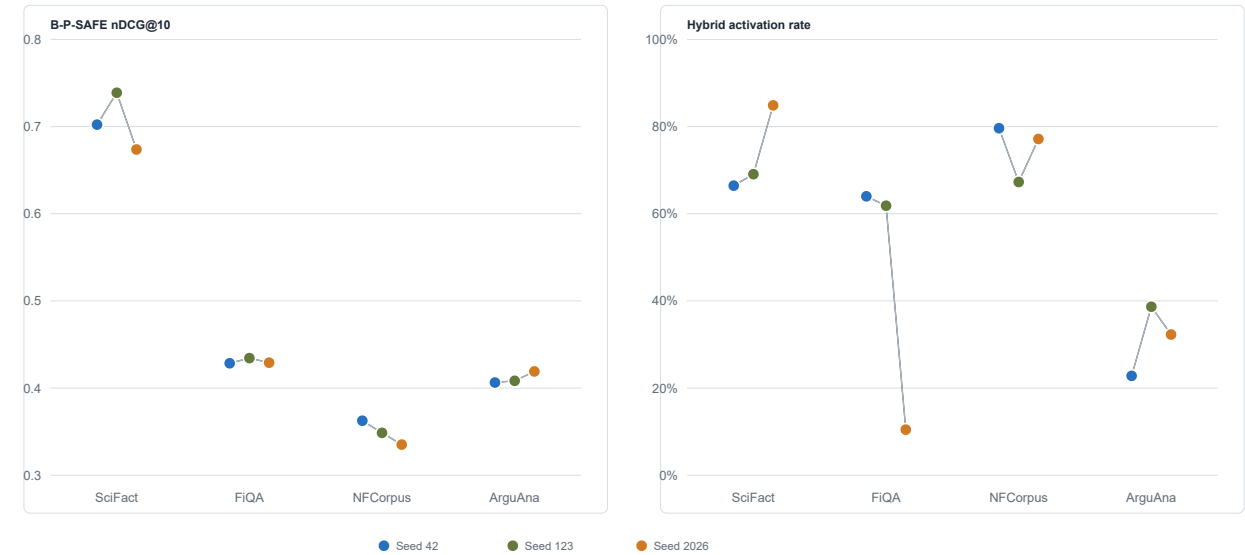


Fig. 6. Per-seed high-recall nDCG@10 and hybrid activation. Lines connect split seeds only as a visual guide; they are not a time series.

APPENDIX A

ROUTER BASELINE TABLES

TABLE IV

ROUTER BASELINE COMPARISON ON SEED 42 (PART I). BASELINE-ROUTER LATENCY EXCLUDES UNINSTRUMENTED DECISION OVERHEAD; B-P-SAFE LATENCY IS MEASURED END-TO-END. THE RANDOM ROUTER USES A VALIDATION-DERIVED ACTIVATION PROBABILITY, NOT MATCHED TEST ACTIVATION.

Dataset	Mode	Method	nDCG@10	Δ_D	Δ_H	Lat. (ms)	LS	HAR	p_D	p_H
SciFact	–	Dense-only	0.6466	+0.0000	-0.0865	3.1	99.6%	0.0%	1.000	2.08×10^{-4}
SciFact	–	Always-Hybrid	0.7330	+0.0865	+0.0000	710.6	0.0%	100.0%	2.08×10^{-4}	1.000
SciFact	–	Random	0.6676	+0.0210	-0.0654	197.2	72.2%	27.6%	0.128	5.80×10^{-4}
SciFact	–	Dense-margin	0.7307	+0.0842	-0.0023	402.7	43.3%	56.6%	4.70×10^{-5}	0.833
SciFact	–	Dense-entropy	0.7132	+0.0667	-0.0198	416.1	41.4%	58.6%	0.001	0.087
SciFact	–	Regression-only	0.6957	+0.0491	-0.0373	441.9	37.8%	61.2%	0.010	0.007
SciFact	–	Classification-only	0.7287	+0.0822	-0.0043	595.8	16.2%	82.2%	3.46×10^{-4}	0.319
SciFact	–	Oracle	0.7663	+0.1197	+0.0333	185.1	73.9%	25.7%	8.85×10^{-9}	2.40×10^{-4}
SciFact	Balanced	B-P-SAFE	0.6965	+0.0499	-0.0365	467.1	34.3%	63.8%	0.011	0.005
SciFact	High recall	B-P-SAFE	0.7023	+0.0557	-0.0308	486.2	31.6%	66.4%	0.006	0.011
FiQA	–	Dense-only	0.4085	+0.0000	-0.0243	5.6	99.4%	0.0%	1.000	0.058
FiQA	–	Always-Hybrid	0.4327	+0.0243	+0.0000	878.3	0.0%	100.0%	0.058	1.000
FiQA	–	Random	0.4211	+0.0127	-0.0116	232.3	73.6%	26.2%	0.124	0.238
FiQA	–	Dense-margin	0.4239	+0.0155	-0.0088	177.0	79.9%	19.7%	0.025	0.415
FiQA	–	Dense-entropy	0.4335	+0.0251	+0.0008	677.9	22.8%	76.9%	0.026	0.902
FiQA	–	Regression-only	0.4373	+0.0289	+0.0046	738.2	16.0%	84.0%	0.010	0.464
FiQA	–	Classification-only	0.4327	+0.0243	+0.0000	872.5	0.7%	99.4%	0.058	1.000
FiQA	–	Oracle	0.4897	+0.0812	+0.0569	286.4	67.4%	31.7%	9.25×10^{-17}	1.03×10^{-14}
FiQA	Balanced	B-P-SAFE	0.4217	+0.0132	-0.0111	499.1	43.2%	45.8%	0.146	0.221
FiQA	High recall	B-P-SAFE	0.4285	+0.0200	-0.0043	632.1	28.0%	64.0%	0.045	0.595

TABLE V

ROUTER BASELINE COMPARISON ON SEED 42 (PART II). BASELINE-ROUTER LATENCY EXCLUDES UNINSTRUMENTED DECISION OVERHEAD; B-P-SAFE LATENCY IS MEASURED END-TO-END. THE RANDOM ROUTER USES A VALIDATION-DERIVED ACTIVATION PROBABILITY, NOT MATCHED TEST ACTIVATION.

Dataset	Mode	Method	nDCG@10	Δ_D	Δ_H	Lat. (ms)	LS	HAR	p_D	p_H
NFCorpus	–	Dense-only	0.3339	+0.0000	-0.0293	2.7	99.6%	0.0%	1.000	0.015
NFCorpus	–	Always-Hybrid	0.3632	+0.0293	+0.0000	721.4	0.0%	100.0%	0.015	1.000
NFCorpus	–	Random	0.3406	+0.0067	-0.0226	233.2	67.7%	31.5%	0.193	0.039
NFCorpus	–	Dense-margin	0.3500	+0.0161	-0.0132	391.1	45.8%	53.7%	0.114	0.045
NFCorpus	–	Dense-entropy	0.3486	+0.0147	-0.0146	294.9	59.1%	38.9%	0.083	0.091
NFCorpus	–	Regression-only	0.3549	+0.0210	-0.0083	197.7	72.6%	28.4%	0.019	0.308
NFCorpus	–	Classification-only	0.3479	+0.0140	-0.0153	265.5	63.2%	37.0%	0.148	0.035
NFCorpus	–	Oracle	0.3941	+0.0602	+0.0309	298.1	58.7%	40.7%	1.10×10^{-9}	2.10×10^{-7}
NFCorpus	Balanced	B-P-SAFE	0.3579	+0.0240	-0.0053	392.7	45.6%	54.9%	0.027	0.314
NFCorpus	High recall	B-P-SAFE	0.3626	+0.0287	-0.0006	574.3	20.4%	79.6%	0.013	0.862
ArguAna	–	Dense-only	0.3946	+0.0000	+0.0030	3.2	99.7%	0.0%	1.000	0.813
ArguAna	–	Always-Hybrid	0.3915	-0.0030	+0.0000	1183.0	0.0%	100.0%	0.813	1.000
ArguAna	–	Random	0.3873	-0.0072	-0.0042	439.8	62.8%	36.8%	0.330	0.687
ArguAna	–	Dense-margin	0.4005	+0.0060	+0.0090	136.0	88.5%	11.2%	0.152	0.458
ArguAna	–	Dense-entropy	0.3920	-0.0026	+0.0004	150.4	87.3%	12.9%	0.549	0.973
ArguAna	–	Regression-only	0.4059	+0.0113	+0.0144	225.7	80.9%	19.4%	0.031	0.218
ArguAna	–	Classification-only	0.4071	+0.0126	+0.0156	398.5	66.3%	33.4%	0.072	0.146
ArguAna	–	Oracle	0.5209	+0.1263	+0.1293	418.3	64.6%	35.3%	7.76×10^{-49}	3.48×10^{-57}
ArguAna	Balanced	B-P-SAFE	0.4069	+0.0123	+0.0154	400.4	66.1%	15.0%	0.008	0.198
ArguAna	High recall	B-P-SAFE	0.4064	+0.0118	+0.0148	469.1	60.3%	22.8%	0.039	0.196

APPENDIX B

PER-SEED DIAGNOSTICS

TABLE VI
PER-SEED ROUTING DIAGNOSTICS USED IN TABLE III.

Dataset	Mode/seed	nDCG@10	LS	HAR	HGR	OGC	RC	HA
SciFact	Balanced / 42	0.6965	34.3%	63.8%	0.577	0.417	0.623	0.003
SciFact	Balanced / 123	0.7356	35.8%	63.2%	0.806	0.545	0.783	0.009
SciFact	Balanced / 2026	0.6674	19.0%	80.9%	0.728	0.367	0.820	0.001
SciFact	High recall / 42	0.7023	31.6%	66.4%	0.644	0.465	0.667	0.003
SciFact	High recall / 123	0.7388	29.9%	69.1%	0.844	0.571	0.848	0.009
SciFact	High recall / 2026	0.6737	14.9%	84.9%	0.862	0.434	0.903	0.001
FiQA	Balanced / 42	0.4217	43.2%	45.8%	0.544	0.163	0.510	0.049
FiQA	Balanced / 123	0.4239	49.9%	35.4%	0.748	0.208	0.495	0.059
FiQA	Balanced / 2026	0.4199	76.2%	4.0%	0.142	0.065	0.075	0.070
FiQA	High recall / 42	0.4285	28.0%	64.0%	0.823	0.246	0.655	0.040
FiQA	High recall / 123	0.4344	28.8%	61.8%	1.299	0.361	0.855	0.044
FiQA	High recall / 2026	0.4291	71.2%	10.5%	0.385	0.177	0.262	0.062
NFCorpus	Balanced / 42	0.3579	45.6%	54.9%	0.819	0.399	0.747	0.012
NFCorpus	Balanced / 123	0.3431	56.1%	42.6%	0.824	0.387	0.537	0.037
NFCorpus	Balanced / 2026	0.3298	29.8%	69.1%	0.827	0.260	0.821	0.020
NFCorpus	High recall / 42	0.3626	20.4%	79.6%	0.978	0.476	0.893	0.006
NFCorpus	High recall / 123	0.3486	32.3%	67.3%	1.036	0.486	0.788	0.029
NFCorpus	High recall / 2026	0.3351	22.4%	77.2%	1.188	0.374	0.938	0.014
ArguAna	Balanced / 42	0.4069	66.1%	15.0%	0.000	0.098	0.190	0.185
ArguAna	Balanced / 123	0.4057	46.6%	39.7%	0.000	0.110	0.460	0.133
ArguAna	Balanced / 2026	0.4190	60.0%	23.7%	0.000	0.158	0.334	0.171
ArguAna	High recall / 42	0.4064	60.3%	22.8%	0.000	0.093	0.258	0.174
ArguAna	High recall / 123	0.4083	47.2%	38.7%	0.000	0.134	0.457	0.139
ArguAna	High recall / 2026	0.4191	53.0%	32.3%	0.000	0.159	0.433	0.160

APPENDIX C

SAVED ROUTER CONFIGURATIONS

TABLE VII
SAVED SEED-42 DEPLOYMENT SETTINGS. LITE IS FIXED; BALANCED AND HIGH-RECALL SETTINGS ARE VALIDATION-SELECTED OR CONFIGURED BEFORE TEST EVALUATION.

Dataset	Mode	τ_{gain}	τ_{harm}	λ_{lat}	λ_{harm}	λ_{rec}	λ_{cand}	LCB flag
SciFact	Lite	0.60	0.20	2.0e-04	0.50	0.10	1.0e-04	on
SciFact	Balanced	0.10	0.20	1.0e-06	0.10	0.25	1.0e-04	off
SciFact	High recall	0.25	0.50	1.0e-05	0.25	0.25	1.0e-05	off
FiQA	Lite	0.60	0.20	2.0e-04	0.50	0.10	1.0e-04	on
FiQA	Balanced	0.25	0.50	1.0e-05	0.25	0.25	1.0e-04	off
FiQA	High recall	0.25	0.50	1.0e-05	0.25	0.25	1.0e-05	off
NFCorpus	Lite	0.60	0.20	2.0e-04	0.50	0.10	1.0e-04	on
NFCorpus	Balanced	0.25	0.50	1.0e-05	0.25	0.25	1.0e-04	off
NFCorpus	High recall	0.25	0.50	1.0e-05	0.25	0.25	1.0e-05	off
ArguAna	Lite	0.60	0.20	2.0e-04	0.50	0.10	1.0e-04	on
ArguAna	Balanced	0.10	0.40	1.0e-06	0.25	0.10	1.0e-04	off
ArguAna	High recall	0.10	0.30	1.0e-06	0.25	0.10	1.0e-05	off

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